

Ninja Locker: A Hand-Gesture-Enabled Knowledge-Based VR Authentication Interface

Isla Xi Han*
Princeton University



Figure 1: *Ninja Locker* 3D user interface (Left: correct gesture combination cued by the purple prompt; Right: wrong gesture combination cued by the blue prompt.)

ABSTRACT

This paper presents *Ninja Locker*, a hand-gesture-enabled knowledge-based virtual reality (VR) 3D authentication interface, in the context of the 2023 3DUI contest at the IEEE VR conference. Unlike most existing biometric-based gestural authentication systems, *Ninja Locker* takes advantage of the information asymmetry between a private VR-glass viewport and outsiders' perspective and creates a knowledge-based authentication process enabled by gestural recognition. The novel addition of confounding gestures also increases the security of the system. The proposed work provides a design framework and a prototype example of the 3DUI technique implementation using Unreal Engine Blueprint on Meta Quest 2 VR Headset.

Index Terms: Security and privacy—Security services—Authentication—; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—; Human-centered computing—Human computer interaction (HCI)—Interaction techniques—Gestural input; Human-centered computing—Interaction design—Interaction design process and methods—User interface design;

1 INTRODUCTION

With the development of image processing and recognition, gestural authentication becomes a possibility. Most hand-gesture authentication methods are based on biometric authentication, where an individual's unique feature is identified [3]. For example, there are skeleton or appearance-based methods (e.g., Kinect-based gesture password [6]), hand tremor-based methods [1], and dynamic biometric hand-gesture authentication systems (e.g., AirAuth [5] and Leap Password [2]).

Unlike the biometric gestural authentication systems mentioned above, this paper presents *Ninja Locker*, a knowledge-based authentication system built upon hand gestures. The method assumes that

*e-mail: xihan@princeton.edu



Figure 2: Four randomly generated 3D icon prompts locate at the front, back, left, and right of the user in the virtual environment



Figure 3: 3D prompts

the VR-glass view-port is not cast to the public. Thus, the authentication method takes advantage of this information asymmetry and cues customize-preset gestures from the users with randomized prompts.

2 METHODOLOGY

Ninja Locker adopts a knowledge-based authentication method where gestures need to match prompts. A working prototype is implemented on a Meta Quest 2 headset using Unreal Engine Blueprint.

2.1 Design Process

2.1.1 Prompt and Password

Using a 4-digit numeric screen lock on a smartphone as a metaphor, *Ninja Locker* instead adopts hand gestures (Fig. 4) as “numbers” and 3D icons (Fig. 3) in four directions (Fig. 2) as prompts. Figure 4 shows a non-extensive list of potential hand gestures. These 23 gestures are selected for demonstration because 1) they can be recognized by Meta Quest 2 Hand Tracking, 2) users can relatively easily make these gestures without painfully twisting fingers, 3) they are less known to be culturally rude.

To set up a password, a user selects one hand (left or right) and one gesture for each of the six 3D prompts (Fig. 3). This process is similar to setting up security questions for password recovery in



Figure 4: Hand Gesture List (non-extensive)

many existing web applications. The 3D prompts can be thought of as questions (or locks), and the custom-set hand gesture as answers (or keys).

2.1.2 Randomized Prompts

One major flaw in many password systems is that if someone around the user can ultimately remember and repeat the user's exact movements, they might be able to unlock the user's account by accurately repeating the movements. In *Ninja Locker*, the randomized prompt feature is designed to conquer this challenge by spawning a random one of the six prompts at each of the four directions every time a user enters that level (map/scene). This feature ensures that the user is making different gestures each time, thus making it difficult to steal the password by simply copy-pasting every movement.

2.1.3 Confounding Gestures

Each prompt is intentionally bound to one hand and one gesture so that the other hand can act as a confounding variable. For example, if the user's unlocking gesture for the prompt "red heart" is left-hand thumbs up, they can choose to pair it with any of the 23 gestures (Fig. 4) for the right hand. Because it is assumed that outsiders cannot see the VR-glass view, the confounding gestures make it exponentially harder to hack *Ninja Locker*.

2.2 Implementation

A 3D user interface (3DUI) prototype of *Ninja Locker* is developed using Unreal Engine 4.27 and Blueprint on the Meta Quest 2 VR headset. For demo purposes, three hand gestures are registered into the program, and free assets¹ from Unreal Engine Marketplace are used as effect placeholders. Each of the six 3D prompts is tagged with a left/right hand and an index of the gesture. Every time a user enters the scene, a random Child Actor of the 3D prompt Blueprint is spawned to four locations surrounding the user. When a user faces a 3D prompt blueprint and makes a hand gesture, the gesture recognition event [4] is triggered to verify the gesture "key". Corresponding visual effects (VFX) will give the user feedback on

¹Basic Pickups VFX Set (Niagara) by Suppart, FX Variety Pack by Kakky



Figure 5: *Ninja Locker* prototype on Meta Quest 2 VR headset

whether they entered the correct gesture. For example, in Figure 1, the green VFX on the left indicates a correct gesture entry, but the red VFX on the right indicates incorrectness.

3 CONCLUSION, LIMITATIONS, AND FUTURE WORK

This paper presents a hand-gesture-enabled knowledge-based VR authentication 3DUI, *Ninja Locker*. Unlike most existing biometric-based gestural authentication systems, *Ninja Locker* uses gesture as an input channel and takes advantage of information asymmetry between users and viewers for VR glasses to form a knowledge-based authentication process. Design features such as randomized 3D prompts and confounding gestures further enhance security.

Limitations and drawbacks of the proposed method include: 1) The system is based on the presumption that the view in the VR glasses is private to the user. Thus, *Ninja Locker* will not work if the casting feature is turned on. 2) VR hand tracking requires sufficient lighting to function. 3) Unlike a set password where people can use vocal recitation or muscle memory to remember, *Ninja Locker* can be more mentally demanding to memorize or execute for users. 4) *Ninja Locker* is limited to devices with hand-tracking capacities. Thus, it can be difficult for users to reset passwords or log in to their accounts on other platforms, such as a 2D website or smartphone app.

The current *Ninja Lockers* prototype focuses on 3DUI design. In the near future, the author aims to have more mock-up features fully developed and run user studies to evaluate usability and security. Data security and corresponding back-end setup are crucial aspects to implement as well. In the longer term, the author envisions combining the *Ninja Lockers* method into multi-modal authentication schemes.

REFERENCES

- [1] M. Atas. Hand tremor based biometric recognition using leap motion device. *IEEE Access*, 5:23320–23326, 2017.
- [2] A. Chahar, S. Yadav, I. Nigam, R. Singh, and M. Vatsa. A leap password based verification system. In *2015 IEEE 7th International Conference on Biometrics Theory, Applications and Systems (BTAS)*, pp. 1–6. IEEE, 2015.
- [3] S. Fong, Y. Zhuang, and I. Fister. A biometric authentication model using hand gesture images. *Biomedical engineering online*, 12(1):1–18, 2013.
- [4] JUST2DEVs. Hand tracking gestures. Available at <https://www.just2devs.com/tutorials>.
- [5] S. Kratz and M. T. I. Aumi. Airauth: a biometric authentication system using in-air hand gestures. In *CHI'14 Extended Abstracts on Human Factors in Computing Systems*, pp. 499–502. 2014.
- [6] M. A. M. Shukran and M. S. B. Ariffin. Kinect-based gesture password recognition. *Australian Journal of Basic and Applied Sciences*, 6(8):492–499, 2012.